

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Science and Analytics

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the concept of Knowledge Discovery from Data (KDD). 7
Describe each phase with examples.
- b) Distinguish between Numeric and Categorical variable. Draw a Histogram for the following data by considering appropriate intervals and frequencies: 8
{10,17,13,23,26,24,29,35,21,45,14,32,38,44,37,39,47}.
2. a) What is empirical distribution? Explain the Power law and Zipf's law with suitable example. 7
- b) What is correlation analysis? From the following data points, 8

X (Studied Hours)	Y (Marks Scored)
2	50
3	60
5	80
6	85
8	90

Calculate the Pearson correlation coefficient between Study Hour and Marks Scored and Interpret the relations.

OR

Following table represents a hypothetical cross-tabulation between "Game Preferences" and "Gender".

	Male	Female	Other
Likes Football	19	23	3
Likes Cricket	75	8	1
Likes Volleyball	3	63	0

Do you think Game Preferences are correlated with Gender? Justify using chi-squared test with significance level 0.05. The critical value for chi-squared statistic with 4 degrees of freedom at 0.05 significance level is 9.4877.

3. a) A real estate analyst collected data from 3 houses in a neighborhood to predict house prices based on two features: square footage (in 100s of square feet) and number of bedrooms. 7

Square footage (X1)	Number of bedrooms (X2)	House price (in \$1000s) (Y)
15	3	240
18	4	290
20	3	310

Find the regression equation: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

- b) Explain R^2 and Adjusted R^2 in the context of model evaluation. What is multicollinearity in multivariate regression? Explain how Variance Inflation Factors (VIFs) help in detecting it. 8
4. a) Why PCA is important? Explain different steps used for computing dimensionality reduction using PCA. 7
- b) Derive Nadaraya-Watson kernel regression estimator. Explain the role of bandwidth in kernel regression and discuss strategies for selecting optimal bandwidth. 8
5. a) Explain k-means clustering algorithm with its advantage and disadvantages. 7
- b) You are given the following dataset of customers, where the target variable is whether the customer purchased a product (Yes or No): 8

Age	Income	Purchased
young	high	no
young	medium	yes
middle	high	yes
senior	Low	no
senior	medium	yes
middle	Low	yes

Using the CART (Classification and Regression Tree) algorithm, construct a binary classification tree (up to two levels deep) for predicting whether a customer will purchase the product.

6. a) What is time series analysis? Explain four components of time series: trend, seasonal variation, cyclical variation and irregular variation. Describe autocorrelation and stationarity with respect to time series analysis. 7

OR

How do you select the optimal lag length in ARIMA model? Discuss the role of Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) in model selection with justification.

- b) Explain the differences between the ARMA (Auto-Regressive Moving Average) and ARIMA (Auto-Regressive Integrated Moving Average) models. Define the components of each model: Auto-Regressive (AR) Moving Average (MA) Integration (I). 8
7. Write short notes on: (Any two) 2×5
- a) Structured and Unstructured data with example
 - b) Statistical Significance: p-value
 - c) Casual Directed Acyclic graph
 - d) Logistic Regression